Orientation Discrimination Task
PSY310 Lab in Psychology
Lab Report

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GitHub Link:

Introduction

Psychophysics, first established by German scientist and philosopher Gustav Theodor Fechner, is a part of psychology that studies the connection between physical stimuli and the perceptual experience in the mind. It dives deep into analyzing how the changes in the physical properties of a stimuli presented change a person's response or experience (Bilal, n.d).

Visual search tasks too have been a prominent way of seeing how attention operates because they require the observer to find a target among competitive distractors under different levels of complexity and difficulty.

The signal detection theory or orientation discrimination to be precise is a part of psychophysics that assesses performance according to the tilt of the stimulus (the gabor patch). This can be related to visual search tasks as well as a baseline of tilt measurement accuracy of individuals.

The task tests for the ability to discriminate orientations in comparison to the absence of the same, which is a vertical condition. It is an experimental method that helps to identify the sensitivity of the individual to identify the signal which is the main stimulus embedded in a vast population of noise which is different from the signal and is the one you need to distinguish (Austin, 2025). The higher the d' sensitivity, the more will be the accuracy to be able to distinguish between the orientations.

The experiment shows two grating manipulations on the participant's screen. It is either vertical/straight (figure 1) which is known as the signal or is tilted (figure 2), known as the noise.

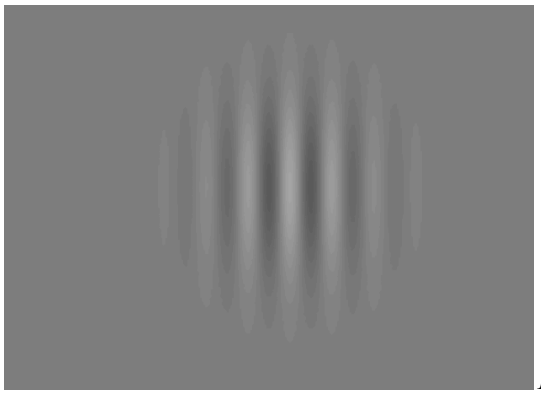


Figure 1: Signal- no tilt

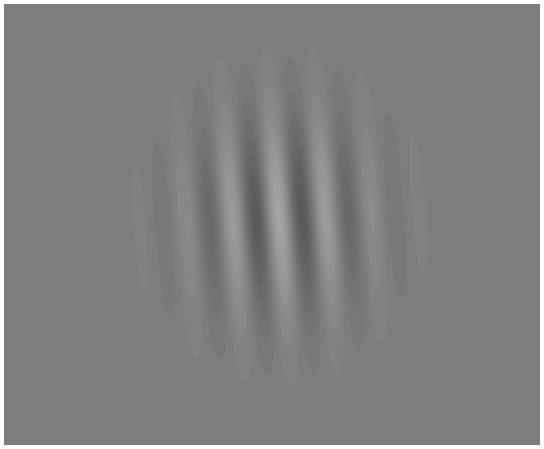


Figure 2: Noise- tilted

Every outcome has 4 possible trials:

1. *Hit:* the signal is present and the participant correctly reports its presence.
2. *Miss*- the signal is present but the participant fails to report it.
3. *False alarm*- the signal is absent but the participant reports it is present.
4. *Correct rejection*- the signal is absent and the participant correctly reports its absence.

In this experiment, the correct responses are hits and correct rejections, and incorrect responses are misses and false alarms (Signal Detection Theory, n.d.).
Also, the way the participant responds tells us about their bias towards answering the questions, in terms of whether they give more preference to one response (yes/no) over the other. A liberal bias is seen when a participant reports the presence of the signal even when they are uncertain, whereas a conservative bias is when the participant tends to be more cautious and hence reports the presence only when they are certain; otherwise, they tend to reject it correctly.

In essence, this experimental design provides information and data of the sensory processing of an individual. It can even be applied in multiple everyday situations or experiments.

including psychologists, scientists, students, etc. For example, measuring the sensitivity of a diagnostician in differentiating between X-rays showing the presence of a tumor and those showing healthy tissue. This helps us identify the sensitivity to signals against a pool of noise, therefore testing their accuracy, perception and decision-making process (Hautus et al., 2021).

Orientation discrimination tasks can be used to investigate the mechanisms and temporal dynamics of visual perception. For example, Mareschal et al. (2006) used a single Gabor stimulus in an orientation discrimination task to examine how visual information supporting orientation judgments changes over time. Their findings suggested that the perceptual template used for orientation discrimination is established within the first 50 ms of stimulus presentation!

Method

Participants

This experiment has one participant who is a 20-year-old female undergraduate 3rd year student of Ahmedabad University. The aim was to assess their accuracy of identifying the tilt of gabor patches shown to her and press the up key if there was no tilt (straight) and the down key if the tilt was to the right or left. Instructions were made clear and consent was received in advance.

Materials and stimuli

The study required a functioning laptop with PsychoPy v2026.2.4 installed. It was conducted on a laptop with a monitor of dimension 13.3”, a resolution of 1920x1080 pixels and a refresh rate of 60 Hz. 100 trials of the experiment were conducted.

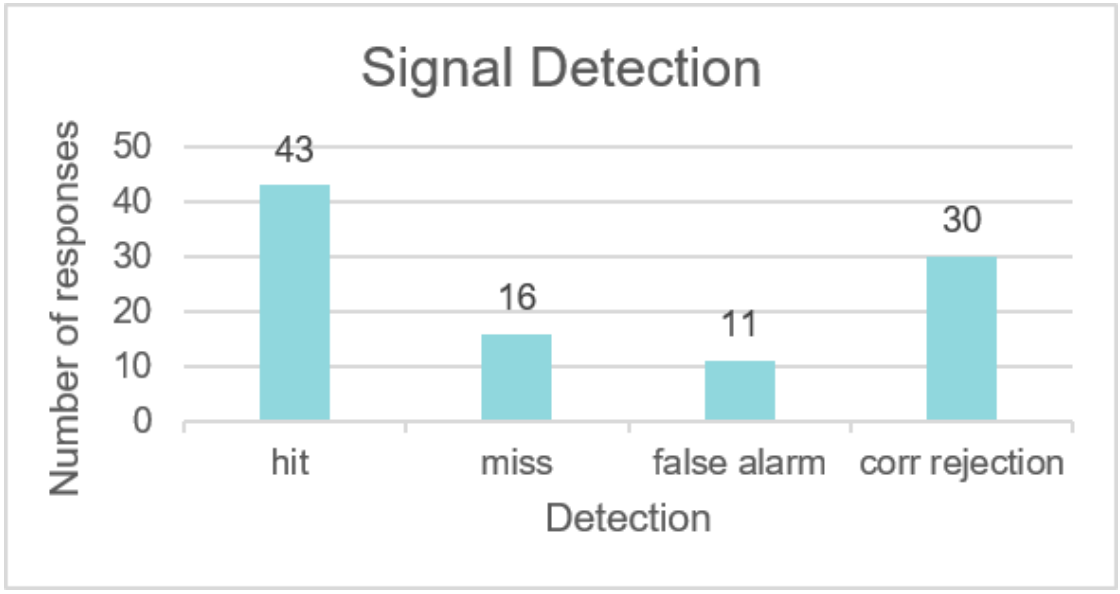
The stimulus was shaped sinusoidal with a Gaussian mask, either vertically oriented or tilted left or right. The contrast was set to 0.3. The size of the stimulus was 0.2 x 0.2. The stimulus examples have been shown in figure 1 and 2 above.

Procedure

The first part of the experiment was an instructions page which instructed the participants their task in the experiment. Then the trials begin with a cross shaped fixation presented for one second. This is followed by the stimulus, a sinusoidal shape with a Gaussian mask (Figure 1 / Figure 2) which appears on the screen for 300 milliseconds. The grating was set to repeat every trial so it randomly changed every trial. This routine was set on loop with a tilt orientation range of [-5,5]. The contrast of 0.3 was set. The participants were required to respond to the straight/vertical grating by pressing on the 'up' [↑] arrow key and respond to the left/right tilt by pressing on the 'down'[↓] arrow key. 100 trials were conducted with the participant.
The experiment was conducted on Psychopy and then put into an Excel sheet in .csv format. Further, the responses were divided into hits, misses, false alarms and correct rejections. The summation of all these outcomes individually was considered as data. This data was used to calculate the sensitivity (d') which is probability of false alarm minus the probability of hit, and. The criterion (C) was calculated too by adding both the probabilities divided by half.

Results

The total of correct responses (hit + correct rejection) is 73 and incorrect responses (false alarm + misses) are 27. The following graph and table show the classification of responses into the 4 categories.



A bar plot representing the number and types of responses over the 100 trials.

direction	Respond YES	Respond NO
signal present (vertical)	43	16
signal absent (tilt)	11	30

The table representing the total outcomes over the 100 trials.

The probability of hits is 0.729 and the probability of false alarms is 0.268. These values were used to calculate the sensitivity (d'), which is 1.227, and criterion (c) which is 0.004.

Calculation Category	Formula	value
Probability of hit	prob hit= hit/hit+ miss	0.728813559
Probability of false alarm	prob FA= FA/FA+ corr rejection	0.268292683
D'	d prime= z(prob hit) - z(prob FA)	1.227213455
C Criterion	c= - (z(prob hit) + z(prob FA))/2	0.004378061

Discussion

The probability of hits was 0.729, indicating that participants correctly identified the absence of a tilt in approximately 72.9% of trials. This is above 50% which means it was a pretty good performance. The probability of false alarms was 0.268, indicating that participants incorrectly reported the presence of signal in approximately 26.8% of trials when a tilt/only noise was actually present. So, the tendency to make errors was moderate. The sensitivity measure (d') was 1.227, suggesting a moderate ability to discriminate between trials with and without a tilt. It was slightly challenging for them.The criterion (c) was 0.004, which is very close to zero, indicating that participants showed little to no response bias and did not consistently favor one response over the other. The average response time across 100 trials was 0.88 milliseconds.

References

1) Bilal. (n.d.). What is Psychophysics? Scribd. https://www.scribd.com/document/515003951/What-is-Psychophysics#:~:text=Psychophysics%20is%20the%20scientific%20study,stimulus%20intensity%20and%20sensation%20magnitude

2) Hautus, M. J., Macmillan, N. A., & Creelman, C. D. (2021). Detection theory. https://doi.org/10.4324/9781003203636

3) Hautus, M. J., Macmillan, N. A., & Creelman, C. D. (2021). Detection theory. https://doi.org/10.4324/9781003203636

4) Heeger, D. (n.d.). *Signal Detection Theory*. Signal detection theory. https://www.cns.nyu.edu/~david/handouts/sdt/sdt.html

5) Mareschal, I., Dakin, S. C., & Bex, P. J. (2006, March 28). *Dynamic Properties of orientation discrimination assessed by using classification images*. Proceedings of the National Academy of Sciences of the United States of America. https://pmc.ncbi.nlm.nih.gov/articles/PMC1458806/